

# CLOUDSHIELD AI

## Software Requirements Specification

*Document 01 | Product-Level Requirements | Version 1.0*

| Field                | Value                                                                   |
|----------------------|-------------------------------------------------------------------------|
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| Status               | Baseline / Pre-Implementation                                           |
| Parent Specification | CSA-00-MSS — CloudShield AI Master System Specification                 |
| Scope                | Entire CloudShield AI product                                           |
| DAIL Position        | Subsystem; this SRS defines product-level DAIL integration requirements |

**Requirement principle: every major capability must be observable, testable, and traceable.**

## Document Control & Contents

| #  | Section                                         |
|----|-------------------------------------------------|
| 1  | 1. Purpose and Scope                            |
| 2  | 2. Product Context                              |
| 3  | 3. Definitions and Requirement Conventions      |
| 4  | 4. Stakeholders, Users and Roles                |
| 5  | 5. Product-Level Functional Requirements        |
| 6  | 6. Cloud Onboarding Requirements                |
| 7  | 7. Asset Discovery & Configuration Requirements |
| 8  | 8. Security Analysis & Finding Requirements     |
| 9  | 9. Risk Intelligence Requirements               |
| 10 | 10. Dependency & Impact Requirements            |
| 11 | 11. AI Intelligence Requirements                |
| 12 | 12. Remediation & Candidate Requirements        |
| 13 | 13. DAIL Integration Requirements               |
| 14 | 14. Verification & Promotion Requirements       |
| 15 | 15. Evidence, Audit & Reporting Requirements    |
| 16 | 16. Dashboard & UX Requirements                 |
| 17 | 17. Experimentation & Evaluation Requirements   |
| 18 | 18. Integration Requirements                    |
| 19 | 19. Security Requirements                       |
| 20 | 20. Privacy & Data Protection Requirements      |
| 21 | 21. Reliability & Resilience Requirements       |
| 22 | 22. Performance & Scalability Requirements      |
| 23 | 23. Observability & Operations Requirements     |
| 24 | 24. Data Requirements                           |
| 25 | 25. API & Inter-Service Requirements            |
| 26 | 26. Configuration & Policy Requirements         |
| 27 | 27. Error Handling Requirements                 |
| 28 | 28. Compatibility & Portability                 |
| 29 | 29. MVP Requirements                            |
| 30 | 30. Traceability & Acceptance                   |
| 31 | 31. Out-of-Scope Requirements                   |
| 32 | 32. Requirements Summary                        |

## 1. Purpose and Scope

This Software Requirements Specification (SRS) defines what CloudShield AI must provide as a complete product. It translates the product boundary established by the Master System Specification into testable functional, data, security, operational, performance, and quality requirements.

The SRS covers CloudShield AI as a whole: cloud onboarding, asset intelligence, configuration analysis, security findings, risk analysis, dependency intelligence, AI assistance, remediation, DAIL lifecycle control, verification, promotion, trusted state, evidence, audit, reporting, experimentation, APIs, integrations, security, reliability, and operations.

Detailed internal DAIL behavior remains in the dedicated DAIL engineering package. This document defines the product-level requirements and interfaces that DAIL must satisfy as a CloudShield AI subsystem.

## 2. Product Context

CloudShield AI operates between authorized cloud environments and human or automated operators. It observes cloud state, normalizes resources into a product model, evaluates security and operational risk, provides contextual AI assistance, and controls infrastructure-changing workflows through deterministic validation and governance.

Primary product control loop:

DISCOVER →INVENTORY →CONFIGURE →ANALYZE →FINDING →RISK →AI EXPLANATION/RECOMMENDATION →  
REMEDIAION CANDIDATE →DAIL IMPACT →VERIFY →DECIDE →PROMOTE →TRUSTED STATE →MONITOR

AI-generated content is never authoritative trusted infrastructure state. It must pass through structured validation and the appropriate lifecycle controls before it can influence infrastructure.

## 3. Definitions and Requirement Conventions

The keywords SHALL/MUST are mandatory; SHOULD is a strong recommendation subject to documented justification; MAY is optional. Requirements must be objectively verifiable through inspection, automated tests, integration tests, or controlled operational validation.

| Term          | Definition                                                                       |
|---------------|----------------------------------------------------------------------------------|
| Asset         | Normalized representation of a cloud resource or relevant infrastructure object. |
| Finding       | Detected security, configuration, policy, or operational issue.                  |
| Candidate     | Structured proposed infrastructure change/state that is not yet trusted.         |
| Trusted State | State accepted after required validation and governance controls.                |
| Verification  | Execution of defined validation checks.                                          |
| Promotion     | Controlled transition of an accepted candidate/result into trusted state.        |
| Evidence      | Artifact or record supporting an observation, action, verification, or decision. |
| DAIL          | Deterministic AI Infrastructure Lifecycle subsystem.                             |
| Invariant     | Condition that must remain true for a state or transition to be valid.           |
| Impact Set    | Resources/components potentially affected by a candidate change.                 |

## 4. Stakeholders, Users and Roles

| Role                      | Primary responsibilities / requirements                                                |
|---------------------------|----------------------------------------------------------------------------------------|
| Security Engineer         | Inspect findings, evidence, risk, dependencies, and remediation candidates.            |
| Cloud / Platform Engineer | Inspect assets/configuration and submit or review infrastructure changes.              |
| Reviewer / Approver       | Approve or reject policy-gated promotions.                                             |
| Administrator             | Manage organizations, projects, environments, access, cloud connections, and policies. |

|                             |                                                                                     |
|-----------------------------|-------------------------------------------------------------------------------------|
| Viewer / Auditor            | Inspect authorized posture, lifecycle history, evidence, and audit records.         |
| CloudShield AI Services     | Execute authorized discovery, analysis, orchestration, verification, and recording. |
| AI Model                    | Provide bounded inference; never becomes the authority for trusted state.           |
| Cloud Provider APIs         | Provide authoritative observations and authorized execution surfaces.               |
| CI/CD / External Automation | Optionally submit candidates, trigger workflows, or consume results.                |

## 5. Product-Level Functional Requirements

| ID     | Requirement                                                                          | Area          |
|--------|--------------------------------------------------------------------------------------|---------------|
| FR-001 | System SHALL support organization, project, and environment boundaries.              | Core          |
| FR-002 | System SHALL authenticate and authorize protected operations.                        | Security      |
| FR-003 | System SHALL support authorized cloud account onboarding.                            | Cloud         |
| FR-004 | System SHALL discover supported cloud resources and normalize them into assets.      | Discovery     |
| FR-005 | System SHALL store versioned configuration observations for supported assets.        | Configuration |
| FR-006 | System SHALL execute defined security/configuration checks and create findings.      | Security      |
| FR-007 | System SHALL associate findings with affected assets and supporting evidence.        | Security      |
| FR-008 | System SHALL calculate or expose a deterministic risk assessment for findings.       | Risk          |
| FR-009 | System SHALL represent relevant asset dependencies and relationships.                | Dependency    |
| FR-010 | System SHALL provide contextual AI explanations using authorized data.               | AI            |
| FR-011 | System SHALL support structured remediation recommendations.                         | AI            |
| FR-012 | System SHALL represent infrastructure-affecting changes as explicit candidates.      | Remediation   |
| FR-013 | System SHALL route infrastructure-changing candidates through DAIL where applicable. | DAIL          |
| FR-014 | System SHALL perform dependency-aware impact analysis where policy requires it.      | DAIL          |
| FR-015 | System SHALL execute required deterministic verification gates.                      | Verification  |
| FR-016 | System SHALL prevent promotion when mandatory verification/policy gates fail.        | Trust         |
| FR-017 | System SHALL record promotion decisions and resulting trusted state.                 | Trust         |
| FR-018 | System SHALL preserve evidence for significant workflow steps.                       | Evidence      |
| FR-019 | System SHALL expose audit history for security/governance-relevant actions.          | Audit         |
| FR-020 | System SHALL provide dashboards and investigation views.                             | UX            |
| FR-021 | System SHOULD support controlled experiments for AI/lifecycle strategies.            | Experiment    |
| FR-022 | System SHALL provide operational health and failure visibility.                      | Operations    |

## 6. Cloud Onboarding Requirements

- FR-101 — The system SHALL allow an authorized administrator to create a cloud connection for a project/environment.
- FR-102 — The system SHALL validate that cloud authorization is usable before marking onboarding successful.
- FR-103 — The system SHALL avoid storing long-lived credentials in ordinary application records when secure references are available.
- FR-104 — The system SHALL identify provider/account context for every cloud observation.

- FR-105 — The system SHALL expose onboarding status and actionable failure information.
- FR-106 — The system SHALL support disabling or revoking a cloud connection.
- FR-107 — Cloud access SHALL use least-privilege permissions appropriate to enabled capabilities.

## 7. Asset Discovery & Configuration Requirements

- FR-201 — Discovery SHALL enumerate supported resource types within authorized scope.
- FR-202 — Every normalized asset SHALL have a stable product identity suitable for correlation across observations.
- FR-203 — Provider identifiers required for reconciliation SHALL be retained.
- FR-204 — Assets SHALL be associated with project, environment, cloud account, resource type, and region/location where applicable.
- FR-205 — Configuration snapshots SHALL be timestamped and versioned.
- FR-206 — Observed configuration SHALL be distinguishable from proposed configuration.
- FR-207 — The system SHALL detect relevant configuration changes between comparable snapshots.
- FR-208 — Partial discovery failures SHALL be explicitly represented and SHALL NOT silently erase unrelated valid state.
- FR-209 — Discovery provenance SHALL identify source and collection time.

## 8. Security Analysis & Finding Requirements

- FR-301 — Security checks SHALL be explicit and versioned.
- FR-302 — Each finding SHALL identify the affected asset and violated condition.
- FR-303 — Findings SHALL include severity or equivalent prioritization information.
- FR-304 — Findings SHOULD include evidence supporting detection.
- FR-305 — The system SHALL support appropriate finding lifecycle states such as open, acknowledged, suppressed, remediated, or resolved.
- FR-306 — Suppression SHALL require authorization and SHALL be auditable.
- FR-307 — Detection evidence SHALL be distinguishable from AI-generated explanation.
- FR-308 — Security analysis SHALL be repeatable against a known configuration snapshot.

## 9. Risk Intelligence Requirements

- FR-401 — The system SHALL produce a risk assessment for supported findings.
- FR-402 — Risk inputs SHALL be traceable to defined signals such as severity, exposure, asset criticality, dependency impact, and policy context.
- FR-403 — Risk scoring SHALL be versioned.
- FR-404 — AI MAY enrich risk explanation but SHALL NOT silently replace deterministic risk inputs.
- FR-405 — Authorized users SHALL be able to inspect why a finding received its priority.

## 10. Dependency & Impact Requirements

- FR-501 — The system SHALL represent explicit dependencies between supported assets/components.
- FR-502 — Dependencies SHALL retain relationship type and provenance where available.
- FR-503 — A candidate SHALL identify resources it directly modifies.
- FR-504 — The system SHALL calculate or invoke impact analysis when policy requires it.
- FR-505 — Impact analysis SHALL identify directly and transitively affected resources where supported.
- FR-506 — Candidates with unresolved high-impact dependencies SHALL be prevented from automatic promotion.
- FR-507 — Authorized users and lifecycle services SHALL be able to query dependency information.

## 11. AI Intelligence Requirements

- FR-601 — The system SHALL provide an AI abstraction layer so model providers can be changed without rewriting product workflows.
- FR-602 — AI requests SHALL be constructed from authorized, task-relevant context.
- FR-603 — Relevant model/provider metadata SHALL be retained for traceability subject to retention/security policy.
- FR-604 — AI explanations SHALL remain grounded in the underlying finding/context and SHALL NOT be treated as authoritative state.
- FR-605 — AI recommendations that may lead to remediation SHALL use a structured representation.
- FR-606 — AI output SHALL pass schema and policy validation before becoming a candidate.
- FR-607 — Malformed, unavailable, unsafe, or otherwise invalid model output SHALL fail safely.
- FR-608 — Raw model output SHALL NOT receive unrestricted cloud credentials or arbitrary production execution authority.
- FR-609 — Users SHALL be able to distinguish AI-generated content from deterministic observations and verification results.
- FR-610 — The system SHOULD support model/prompt/version evaluation through experimentation.

## 12. Remediation & Candidate Requirements

- FR-701 — Every infrastructure-affecting remediation SHALL be represented as an explicit candidate.
- FR-702 — Candidates SHALL include source, target scope, proposed changes, reason, and creation metadata.
- FR-703 — Candidate revisions SHALL be versioned and traceable.
- FR-704 — Candidate structure SHALL be validated before lifecycle processing.
- FR-705 — Candidates SHALL retain links to motivating findings, assets, and relevant evidence/context.
- FR-706 — Candidate rejection SHALL support an auditable reason.
- FR-707 — The system SHALL support candidate supersession when a newer candidate replaces an older one.

## 13. DAIL Integration Requirements

- FR-801 — CloudShield AI SHALL provide DAIL with structured candidate/context through an explicit interface.
- FR-802 — DAIL SHALL return lifecycle state and decision-relevant outputs through a defined contract.
- FR-803 — CloudShield AI SHALL treat DAIL verification/promotion results as lifecycle evidence, not AI prose.
- FR-804 — DAIL SHALL be isolated from unrelated product concerns through explicit module/service boundaries.
- FR-805 — Product workflows SHALL expose DAIL lifecycle status without requiring users to understand internal implementation.
- FR-806 — DAIL failure SHALL produce explicit non-success states and SHALL NOT be interpreted as successful promotion.
- FR-807 — Product candidate history SHALL remain traceable to the DAIL candidate/version.

## 14. Verification & Promotion Requirements

- FR-901 — The system SHALL define verification gates for each promotion class.
- FR-902 — Verification SHALL operate against an identified candidate/version and relevant environment/state.
- FR-903 — Verification results SHALL include timestamp, checks, outcome, and evidence references.
- FR-904 — Failed mandatory checks SHALL block promotion.
- FR-905 — Verification SHALL be repeatable where technically feasible.
- FR-906 — Promotion SHALL require an explicit policy decision or authorized approval path.
- FR-907 — Trusted-state records SHALL be created only after successful promotion.
- FR-908 — Promotion SHALL be idempotent or safely detectable as already completed.
- FR-909 — Rollback/recovery behavior SHALL be defined for supported high-risk changes.

## 15. Evidence, Audit & Reporting Requirements

- FR-1001 — Significant observations, actions, decisions, and lifecycle transitions SHALL generate evidence or audit records.
- FR-1002 — Evidence SHALL link to relevant asset, candidate, verification, or decision.
- FR-1003 — Audit records SHALL identify actor/service, action, target, time, and result.
- FR-1004 — Ordinary users SHALL NOT be able to silently alter historical audit records.
- FR-1005 — Authorized users SHALL be able to inspect candidate and promotion history.
- FR-1006 — Reports SHALL distinguish observed, proposed, verified, and trusted state.
- FR-1007 — Exported reports SHOULD include identifiers enabling traceability to evidence.

## 16. Dashboard & UX Requirements

- UX-001 — The primary dashboard SHALL summarize posture, high-priority findings, active remediation, lifecycle state, and system health.
- UX-002 — Users SHALL navigate from a finding to its affected asset and supporting evidence.
- UX-003 — Asset views SHALL expose configuration, relationships, findings, and lifecycle history.
- UX-004 — Remediation views SHALL distinguish proposed changes from deployed/trusted state.
- UX-005 — AI-generated explanations SHALL be visually distinguishable from authoritative observations.
- UX-006 — Candidate review SHALL expose impact, verification status, policy requirements, and evidence before promotion.
- UX-007 — Promotion history SHALL show authorization and verification outcome.
- UX-008 — Error states SHALL explain failure category and next safe action where possible.

## 17. Experimentation & Evaluation Requirements

- FR-1101 — The system SHOULD support controlled trials of AI models, prompts, remediation strategies, or lifecycle policies.
- FR-1102 — Experiment inputs/results SHALL be isolated from trusted production state.
- FR-1103 — Experiment records SHALL identify strategy/model/version and evaluation metrics.
- FR-1104 — Experiments SHALL NOT bypass production verification or promotion controls.
- FR-1105 — AI evaluation SHOULD measure correctness, safety, usefulness, and failure modes, not only response quality.

## 18. Integration Requirements

| Integration              | Requirement                                                                                     |
|--------------------------|-------------------------------------------------------------------------------------------------|
| AWS / Cloud Provider     | MUST use authenticated least-privilege APIs and preserve provider identifiers.                  |
| LLM Provider             | MUST be accessed through an abstraction boundary with timeout/error handling.                   |
| Source Control           | SHOULD support traceable candidate/configuration references when enabled.                       |
| CI/CD                    | SHOULD support verification triggers/results when enabled.                                      |
| Identity Provider        | SHOULD support enterprise authentication integration.                                           |
| Notification / Ticketing | MAY receive findings, decisions, or lifecycle events.                                           |
| Webhooks / API Clients   | MUST use authentication, authorization, validation, and replay-safe semantics where applicable. |

## 19. Security Requirements

- SEC-001 — Protected APIs SHALL enforce authentication and authorization.
- SEC-002 — Authorization SHALL respect organization/project/environment boundaries.

- SEC-003 — Cloud credentials SHALL use secure storage/references and SHALL not be unnecessarily exposed to users or models.
- SEC-004 — Cloud and internal services SHALL follow least privilege.
- SEC-005 — Infrastructure-affecting operations SHALL require explicit authorization.
- SEC-006 — Raw AI output SHALL never be sufficient authorization for privileged cloud actions.
- SEC-007 — The AI architecture SHALL address prompt/context injection and tool-abuse risks.
- SEC-008 — Sensitive data SHALL be minimized in prompts and logs.
- SEC-009 — Privileged audit events SHALL be protected from unauthorized modification.
- SEC-010 — Tenant isolation failures SHALL be treated as critical security defects.
- SEC-011 — External input SHALL be validated before persistence or execution.
- SEC-012 — Supported production changes SHALL have a recoverable operational path.

## 20. Privacy & Data Protection Requirements

- PRIV-001 — The system SHALL collect only data necessary for enabled capabilities.
- PRIV-002 — Data access SHALL follow least privilege.
- PRIV-003 — Sensitive values SHALL be redacted from ordinary logs and AI context where possible.
- PRIV-004 — Retention periods SHALL be configurable for applicable evidence and AI artifacts.
- PRIV-005 — Data deletion/export behavior SHALL be defined for applicable organizational data.
- PRIV-006 — External model transmission SHALL be explicit and policy-controlled for sensitive deployments.

## 21. Reliability & Resilience Requirements

- REL-001 — Failure of a non-critical discovery target SHALL not silently erase valid previously known state.
- REL-002 — Workflow operations SHALL be idempotent or deduplicated where retries are possible.
- REL-003 — Transient external failures SHALL use bounded retry/backoff.
- REL-004 — Critical state transitions SHALL be persisted transactionally or through an equivalent consistency mechanism.
- REL-005 — Promotion failures SHALL leave the last trusted state identifiable.
- REL-006 — Recovery procedures SHALL be documented and tested for critical services/data.
- REL-007 — Mandatory safety gates SHALL fail closed.

## 22. Performance & Scalability Requirements

| ID       | Requirement                                                            | Initial target                               |
|----------|------------------------------------------------------------------------|----------------------------------------------|
| PERF-001 | Interactive read APIs SHOULD meet a p95 latency target.                | ≤ 500 ms for normal cached/simple reads      |
| PERF-002 | Standard application writes SHOULD be responsive.                      | ≤ 1 s p95 excluding long external operations |
| PERF-003 | Discovery and verification SHALL support asynchronous execution.       | Required                                     |
| PERF-004 | UI SHALL show progress/state for asynchronous workflows.               | Required                                     |
| PERF-005 | Stateless services SHOULD support horizontal scaling.                  | Required                                     |
| PERF-006 | AI calls SHALL use bounded timeouts and concurrency controls.          | Required                                     |
| PERF-007 | Targets SHALL be re-baselined from MVP measurements before production. | Required                                     |

## 23. Observability & Operations Requirements

- OPS-001 — Services SHALL emit structured logs with correlation identifiers.



- OPS-002 — Critical workflows SHALL expose success, failure, latency, and backlog metrics.
- OPS-003 — Distributed workflows SHOULD support tracing across API, AI, DAIL, and cloud boundaries.
- OPS-004 — Service and dependency health SHALL be observable.
- OPS-005 — Security-relevant failures SHALL be distinguishable from ordinary operational failures.
- OPS-006 — Alerts SHALL be actionable and tied to defined thresholds.
- OPS-007 — Deployment/version metadata SHALL be traceable for lifecycle decisions.

## 24. Data Requirements

The product data model SHALL support at minimum:

- Organization, Project, Environment, and CloudAccount.
- Resource/Asset and provider identity.
- Observed configuration snapshots and proposed configuration.
- Findings and risk assessments.
- Dependency relationships.
- Candidates and candidate versions.
- Verification results.
- Trusted state.
- Evidence and audit events.
- Experiments/trials.
- Users, roles, and policies.

Each authoritative entity SHALL have defined ownership, identifiers, lifecycle, timestamps, and access-control expectations in the Domain & Data Model specification.

## 25. API & Inter-Service Requirements

- API-001 — APIs SHALL use explicit versioned contracts where compatibility matters.
- API-002 — APIs SHALL validate input schemas.
- API-003 — APIs SHALL return stable error categories/codes sufficient for client handling.
- API-004 — Protected endpoints SHALL enforce authorization at the resource boundary.
- API-005 — Mutating operations SHALL support idempotency or safe retry semantics where relevant.
- API-006 — Long-running workflows SHALL expose asynchronous job/status semantics.
- API-007 — Service-to-service calls SHALL authenticate and authorize the calling service.
- API-008 — AI and DAIL controls SHALL use structured schemas rather than free-form prose as the primary interface.

## 26. Configuration & Policy Requirements

- POL-001 — Security rules/policies SHALL be versioned.
- POL-002 — Promotion policies SHALL define required verification and approval conditions.
- POL-003 — Policy changes SHALL be auditable.
- POL-004 — Environment-specific policy SHALL be possible subject to inherited organization/project controls.
- POL-005 — Policy evaluation results SHALL identify the policy version used.
- POL-006 — Default safety behavior SHALL deny promotion when required policy evaluation cannot be completed.

## 27. Error Handling Requirements

- ERR-001 — User-visible errors SHALL distinguish validation, authorization, external dependency, processing, and safety failures where possible.

- ERR-002 — Internal errors SHALL not disclose secrets or sensitive implementation details.
- ERR-003 — Failed AI inference SHALL degrade to a safe non-AI path where feasible.
- ERR-004 — Failed verification SHALL never be represented as success.
- ERR-005 — Partial discovery SHALL identify incomplete scope.
- ERR-006 — Retry/recovery SHALL not duplicate infrastructure mutations.

## 28. Compatibility & Portability

- COMP-001 — Core product services SHOULD remain cloud-provider-independent where the domain permits.
- COMP-002 — Provider-specific behavior SHALL be isolated behind integration boundaries.
- COMP-003 — Provider-specific model APIs SHALL be isolated behind the AI abstraction layer.
- COMP-004 — Infrastructure definitions SHALL be version controlled.
- COMP-005 — MVP SHALL prioritize one provider deeply rather than shallow multi-cloud support.

## 29. MVP Requirements

- One primary cloud integration, initially AWS.
- Controlled AWS onboarding.
- Discovery for a focused set of useful resource types.
- Versioned configuration snapshots.
- Focused security checks and actionable findings.
- Deterministic baseline risk scoring.
- AI explanations and structured remediation proposals.
- Explicit remediation candidates.
- DAIL identity/dependency/impact lifecycle integration.
- Bounded deterministic verification.
- Policy-controlled promotion to trusted state.
- Evidence and audit trail.
- Dashboard covering discovery, findings, remediation, lifecycle, and evidence.
- Basic operational monitoring and failure visibility.

MVP acceptance SHALL demonstrate the complete loop on a bounded scenario, not merely isolated feature demonstrations.

## 30. Traceability & Acceptance

Every requirement SHALL be traceable to one or more architecture components, functional workflows, implementation tasks, and tests. A requirement that cannot be objectively verified must be refined before implementation.

| Traceability layer  | Required artifact                           |
|---------------------|---------------------------------------------|
| Product             | 00 — Master System Specification            |
| Requirements        | 01 — This SRS                               |
| Architecture        | 02 — System Architecture Specification      |
| Behavior            | 03 — Functional Specification               |
| UX                  | 04 — Product & UX Specification             |
| Data                | 05 — Domain & Data Model                    |
| Lifecycle           | 06 — State & Lifecycle                      |
| Identity/Dependency | 07 — Identity & Dependency                  |
| Security            | 08 — Security & Threat Model                |
| AI                  | 09–11 — AI Architecture / LLM / Guardrails  |
| DAIL                | DAIL subsystem engineering package          |
| Infrastructure      | AWS & Terraform Specification               |
| Validation          | System Testing / AI Evaluation              |
| Delivery            | Implementation Roadmap / Definition of Done |

## 31. Out-of-Scope Requirements

- Training a foundation model from scratch is not an MVP requirement.
- Building a proprietary local foundation model is not an MVP requirement.
- Universal autonomous production remediation is not an MVP requirement.
- Full multi-cloud parity is not an MVP requirement.
- Replacing all enterprise security tooling is not an MVP requirement.
- Unbounded AI-agent access to cloud credentials is explicitly excluded.

## 32. Requirements Summary

CloudShield AI must provide a secure, observable, testable product that connects cloud discovery and security intelligence to AI-assisted remediation and a deterministic infrastructure lifecycle. The product must preserve the distinction between observed cloud state, proposed state, verified results, and trusted state.

The central product requirement is controlled progression:

OBSERVE → UNDERSTAND → PROPOSE → ANALYZE IMPACT → VERIFY → GOVERN → PROMOTE → RECORD → MONITOR

**Architectural requirement: AI may accelerate reasoning and generation, but it must not become the authority for trusted infrastructure state.**